

Name: _____

These questions assess your understanding of the material from Chapters 9, 10 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

- Grace is with her daughter, Linda, at a park. She is continuously pushing on a 53-kg merry-go-round with a force of 119 N at its edge, 1.5 m away from the center. Linda watches excitedly. Calculate the angular acceleration that is produced in the merry-go-round when no one is on it. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: _____

- Kevin places an outdoor sandwich-board advertising sign on the ground. The sign used to have a chain 0.21 m vertically below its hinge, but the chain is broken. The sign's mass is 16 kg. The sign has a vertical height of 0.79 m. The hinge is opened so that the feet of the sign measure 1.3 m apart. What is the minimum coefficient of static friction between the feet of the sign and the sidewalk? To draw the sign, you may draw a large A. The chain, now broken, would have resembled the horizontal line in the A and the hinge is at the top of the A.

ANSWER: _____

- Michael places a ladder of mass 17 kg against his house. The feet of the ladder are resting on his concrete patio at a distance of 0.71 m from the house. Between the ladder and the concrete, $\mu_s = 0.42$. The top of the ladder rests against a plastic rain gutter. The friction between the ladder and gutter is negligible. The top of the ladder is at a height of 1.14 m above the concrete patio. The ladder's center of gravity is at its midpoint. Michael is standing on a rung at a horizontal distance of 0.21 m from the house. What is the maximum mass that Michael could have without causing the ladder to slip on the concrete?

ANSWER: _____

- Ryan is designing a new 2.5-m-long, 22-kg diving board for DiveMaster, Inc. He bolted a prototype into the ground 2.2 m away from the diving edge with its spring being 1.8 m away from the diving edge. Ryan notes that the spring compresses by 4 cm at equilibrium under the weight of the diving board alone. When Ryan, who has a mass of 73 kg, gets on the prototype diving board at a distance of 0.9 m from the diving edge, by how much does the spring compress at equilibrium? Report your answer in centimeters.

ANSWER: _____

- Alexis is designing a set to be shown in a play. A 81-kg actor stands on a 119-kg platform that is suspended by two ropes, one on its left side, and another on its right side. The left rope is attached to a weight of mass m_L through a frictionless pulley; the right rope is attached to a weight of mass m_R through another frictionless pulley. When the weights are released, the platform must descend with an acceleration of 0.8 m/s^2 without rotating. Alexis needs to make the platform 2.5-m long and the actor must stand 1.1-m from the left side. What must m_L be?

ANSWER: _____

- William is building a seesaw for his daughter, Isabella, and plans on making it 17 ft long. He plans to sit on one side, 6.8 ft away from the center of the seesaw, and his daughter will sit on the other side, 6.8 ft away from the center of the seesaw. William has weighed himself, his daughter, and the seesaw and learned that his weight is 230 lbs, his daughter's weight is 65 lbs, and the weight of the seesaw is 14.8 lbs. He wants to place the fulcrum so that the seesaw is perfectly balanced, so that Isabella can cause a rotation in the seesaw just as easily as himself. Where must the fulcrum be placed?

ANSWER: _____

- Nicholas, a construction worker, holds up a plank of wood 2.6-m long. His left hand pushes an end of the plank down with a force F_L , while his right hand pushes the plank up with a force F_R at a distance of 0.3 m from the left hand. The plank is held in equilibrium. The plank has a mass of 12 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_L .

ANSWER: _____

- Two children, Elizabeth and William, and a toy train are balancing on a long seesaw of negligible mass. Elizabeth has a mass of 24 kg and sits 3.2 m away from the fulcrum. William, who has a mass of 15 kg, sits on the other side of the fulcrum, opposite to Elizabeth. The toy train has a mass of 0.9 kg and is on William's side of the seesaw, a distance of 0.9 m away from the fulcrum. What is William's distance from the fulcrum?

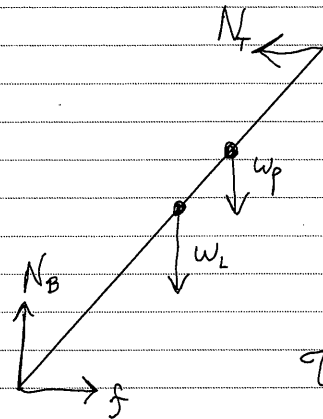
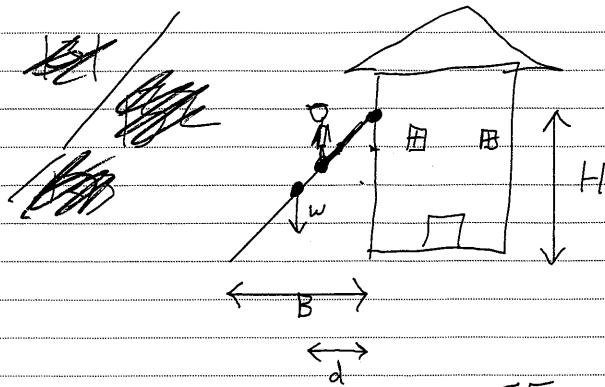
ANSWER: _____

9. Andrew flips his bicycle upside down and moves the pedals so that the rear wheel begins spinning. If the wheel starts from rest and reaches a final angular velocity of 276 rpm in 4.8 s, calculate the angular acceleration in rad/s^2 .
ANSWER: _____
10. A typical small rescue helicopter has four blades, each of which is 3.9-m long and has a mass of 52 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1240 kg. If the helicopter's blades rotate at 283 rpm on the ground, to what height could the helicopter rise if all of its blades' rotational kinetic energy is used to lift it?
ANSWER: _____
11. Sean is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. How much work does he do if he exerts a force of 248 N through a rotation of 0.87 rad (50°)? He exerts the force perpendicular to the grindstone's 39-cm radius and the effects of friction are negligible. Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.
ANSWER: _____
12. Kyle wants to place three blocks on a 2.4-kg, 15-m-long wooden board to make it balance atop a triangular wedge of wood. He places a 40-kg purple block a distance of 0.72 m from the left end of the board, a 2.5-kg green block a distance of 12.5 m from the right end of the board, and places the board on the wedge at a distance of 1.6 m from the left end of the board. Where must he place a 2.8-kg red block, measured from the right end of the board, in order to balance the board?
ANSWER: _____
13. Julia is with her daughter, Grace, at a park. Grace has climbed onto a merry-go-round. Julia is continuously pushing on the 24-kg merry-go-round with a force of 86 N at its edge, 1.3 m away from the center. Calculate the angular acceleration that is produced in the merry-go-round while 34-kg Grace is 0.26 m away from its center. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.
ANSWER: _____
14. Nicholas is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. He exerts a force of 378 N through a rotation of 1.43 rad (82°). He exerts the force perpendicular to the grindstone's 59-cm radius and the effects of friction are negligible. What is the grindstone's final angular velocity if its mass is 88 kg? Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.
ANSWER: _____
15. Robert flips his bicycle upside down and moves the pedals so that the rear wheel begins spinning. The wheel starts from rest and reaches a final angular velocity of 292 rpm in 6.7 s. Robert then slams on the brakes to cause an angular acceleration of -40 rad/s^2 . How long does it take the wheel to stop?
ANSWER: _____
16. A typical small rescue helicopter has four blades, each of which is 3.8-m long and has a mass of 49 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 870 kg. Calculate the rotational kinetic energy that the helicopter possesses when its blades rotate at 347 rpm.
ANSWER: _____
17. Sean, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.1 cm from the reel's axis of rotation. The reel is given an angular acceleration of 122 rad/s^2 for 1.3 s. How many meters of fishing line come off the reel in this time?
ANSWER: _____

PHY201 Practice Exam on chapter(s): 9, 10

1. 2.99 rad/s^2
2. 0.41
3. 99.27 kg
4. 22.2 cm
5. 81.85 kg
6. 3.6 ft from the center, on William's side
7. 392.0 N
8. 5.1 m
9. 6.02 rad/s^2
10. 38.11 m
11. 84.1 J
12. 6.7 m
13. 4.95 rad/s^2
14. 6.45 rad/s
15. 0.76 s
16. 622.9 kJ
17. 4.23 m

#3



$$\sum F_y = 0$$

$$N_B = W_L + W_P$$

$$\sum F_x = 0$$

$$f = N_T$$

$$\sum \tau = 0 \quad \text{say pivot point is at bottom.}$$

$$\tau_{N_B} + \tau_f + \tau_{W_L} + \tau_{W_P} + \tau_{N_T} = 0$$

$$0 + 0 - \tau_L W_L - \tau_L W_P + \tau_L N_T = 0$$

$$- \left(\frac{B}{2} \right) W_L - (B-d) W_P + H f = 0$$

$$f = \frac{1}{H} \left(\frac{B}{2} W_L + (B-d) W_P \right)$$

$$f = \frac{g}{H} \left(\frac{B m_L}{2} + (B-d) m_P \right)$$

$$\mu_s \geq \left(\frac{g}{m_L g + m_P g} \right) \frac{1}{H} \left(\frac{B m_L}{2} + (B-d) m_P \right)$$

$$\mu_{s \min} = \left(\frac{1}{m_L + m_P} \right) \left(\frac{1}{H} \right) \left(\frac{B m_L}{2} + (B-d) m_P \right)$$

133 Max m_P given μ_s

$$f_s = \frac{1}{H} \left(\frac{B}{2} W_L + (B-d) W_P \right)$$

$$f_s \leq \mu_s N_B$$

$$f_s \leq \mu_s (W_L + W_P)$$

$$\mu_s g (m_L + m_P) \geq \frac{g}{H} \left(\frac{B}{2} m_L + (B-d) m_P \right) \quad \text{! (to prevent slipping)}$$

$$\mu_s m_L + \mu_s m_P \geq \frac{B}{2H} m_L + \frac{B-d}{H} m_P$$

$$\left(\mu_s - \frac{B}{2H} \right) m_L \geq \left(\frac{B-d}{H} - \mu_s \right) m_P$$

$$m_P \leq m_L \left(\frac{\mu_s - \frac{B}{2H}}{\left(\frac{B-d}{H} \right) - \mu_s} \right)$$

note

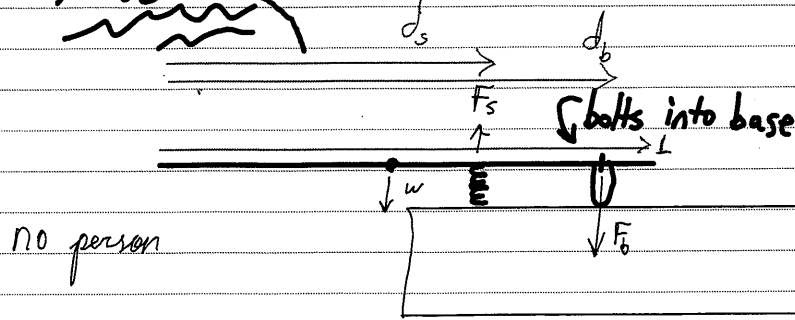
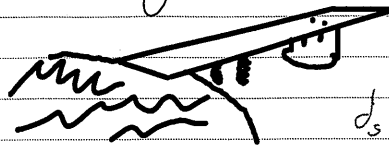
$$x \leq \frac{a-b}{c-d}$$

$$x \geq \frac{b-a}{d-c}$$

division alone by a negative
flips the sign

$$m_{P \max} = m_L \left(\frac{\mu_s - \frac{B}{2H}}{\left(\frac{B-d}{H} \right) - \mu_s} \right)$$

#4

~~#3~~ Diving Board

$$a_y \therefore \sum F_y = 0 \therefore F_s = w + F_b$$

$$F_b = K\Delta x - mg$$

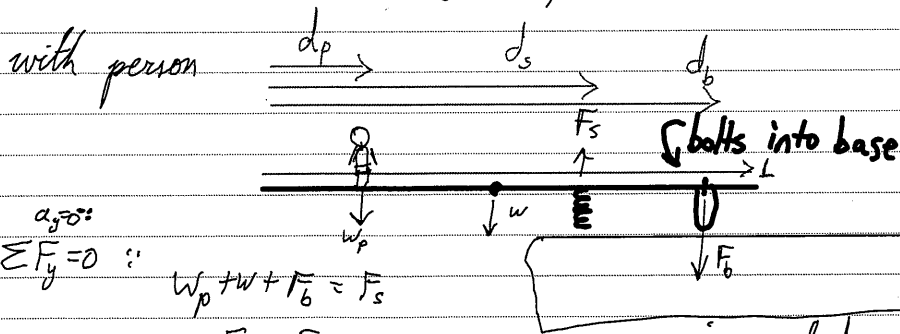
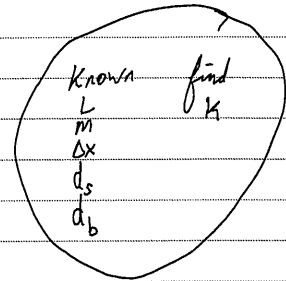
$$(pp \text{ at left}) \quad \sum \tau = 0 \quad \tau_w + \tau_s + \tau_b = 0$$

$$-\frac{L}{2}w + d_s F_s - d_b F_b = 0$$

$$-\frac{L}{2}mg + d_s K\Delta x - d_b K\Delta x + d_b mg = 0$$

$$K\Delta x(d_s - d_b) = mg\left(\frac{L}{2} - d_b\right)$$

$$K = \frac{mg\left(\frac{L}{2} - d_b\right)}{\Delta x_0(d_s - d_b)}$$



$$a_y = 0 \therefore \sum F_y = 0 \therefore w_p + w + F_b = F_s$$

$$pp \text{ at left} \quad F_b = F_s - w_p - w$$

$$\sum \tau = 0 \quad \tau_p + \tau_w + \tau_s + \tau_b = 0$$

$$-d_p w_p - \frac{L}{2}w + d_s F_s - d_b F_b = 0$$

$$-d_p w_p - \frac{L}{2}w + d_s F_s - d_b F_s + d_b w_p + d_b w = 0$$

$$w_p(d_b - d_p) + w(d_b - \frac{L}{2}) - F_s(d_b - d_s) = 0$$

$$\Delta x_p = \frac{w_p(d_b - d_p) + w(d_b - \frac{L}{2})}{K(d_b - d_s)} = \frac{\Delta x_0}{w(d_b - \frac{L}{2})} \left(w_p(d_b - d_p) + w(d_b - \frac{L}{2}) \right)$$

$$\Delta x_p = \Delta x_0 \left(\frac{m_p(d_b - d_p)}{m(d_b - \frac{L}{2})} + 1 \right)$$

given m_p & d_p
(& L, m, d_s, d_b)
find Δx
found K already
above

#5

~~31/35~~

$$\textcircled{1} \Sigma F_y = ma$$

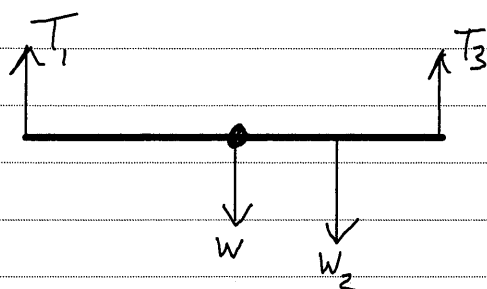
$$T_1 - m_1 g = m_1 a$$

$$T_1 = m_1 (a + g)$$

$$\textcircled{3} \Sigma F_y = m_3 a$$

$$T_3 - m_3 g = m_3 a$$

$$T_3 = m_3 (a + g)$$



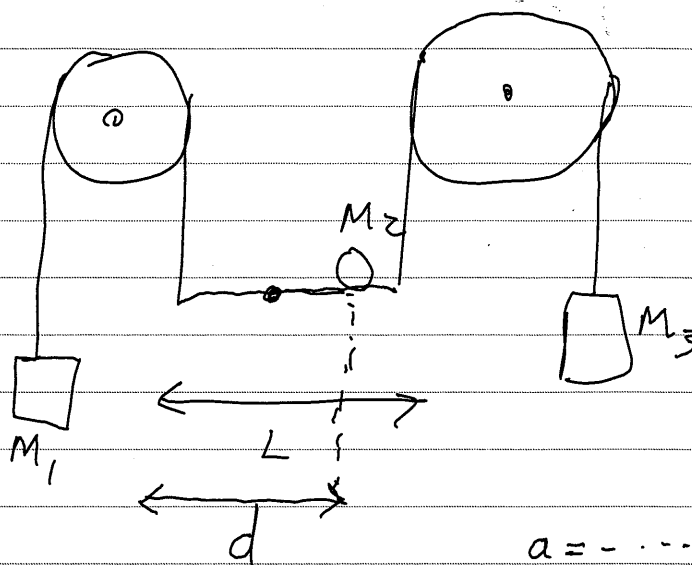
$$\Sigma \tau = 0 \quad \text{pp at left}$$

$$\tau_1 + \tau_w + \tau_2 + \tau_3 = 0$$

$$0 - \frac{L}{2}(mg) - d(m_2 g) + L(T_3) = 0$$

$$L T_3 = g \left(\frac{L}{2} m + d m_2 \right)$$

$$L m_3 (a + g) = g \left(\frac{L}{2} m + d m_2 \right)$$



$$a = - \dots a \downarrow$$

$$\Sigma F_y = -m \ddot{a} - m_2 \ddot{a}$$

$$T_1 + T_3 - mg - m_2 g = ma - m_2 a$$

$$T_1 + T_3 - m(g - a) - m_2(g - a) = 0$$

$$m_1(a + g) + m_3(a + g) - m(g - a) - m_2(g - a) = 0$$

m_1 & m_3 unknown

$$(m_1 + m_3)(a + g) - (g - a)(m + m_2) = 0$$

$$m_1 + m_3 = \frac{(g - a)(m + m_2)}{a + g}$$

$$m_1 = \frac{(g - a)}{a + g} (m + m_2) - m_3$$

$$m_3 = \left(\frac{g \left(\frac{L}{2} m + d m_2 \right)}{L(a + g)} \right)$$

$$\therefore m_1 = \left(\frac{1}{a + g} \right) \left((g - a)(m + m_2) - \left(\frac{g}{L} \right) \left(\frac{L}{2} m + d m_2 \right) \right)$$